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Convolutional neural network (CNN) and federated learning-based privacy preserving approach for skin disease classification

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Abstract
This research displays inspect a study on the classification of human skin diseases using medical imaging, with a focus on data privacy preservation. Skin disease diagnosis is primarily done visually and can be challenging due to variant colors and complex formation of diseases. The proposed solution involves an image dataset with seven classes of skin disease, a convolutional neural network (CNN) model, and image augmentation to increase dataset size and model generalization. The suggested CNN model attained an average precision of 86% and an average recall of 81% for all seven classes of skin diseases. To safeguard the privacy of the data, a federated learning method was used, in which the information was split among 500, 1000, and 2000 users. With the proposed scheme which based on CNN for disease classification and the federated learning method, the average accuracy was 82.42%, 87.26%, and 93.25% for the different numbers of clients. The findings show that it may be possible to effectively categorize skin illnesses by employing a CNN-based approach coupled with federated learning in order to achieve this goal. This would be conducted without compromising the confidentiality of patient data.

Keywords Skin disease classification · Medical imaging · Convolutional neural network (CNN) · Federated learning · Data privacy · Data security · Performance evaluation

1 Introduction

Skin diseases are a widespread and significant health concern globally, with various factors influencing their impact, including environmental factors and genetic susceptibility [1]. Also, social factors like socioeconomic status, schooling, free

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time, and getting to healthcare play a part in how prevalent skin illnesses are and how bad they are [2]. As per the study in [3], skin disease were ranked fourth in the catalog of most plebeian nonfatal diseases globally. Skin diseases can lead to psychological and sociological issues, such as depression, anger, anxiety, low self-esteem and even social isolation [4]. Proper identification is crucial for effective therapy, but doctors have trouble making accurate diagnoses because many skin diseases look identical in shape and color [5]. Machine learning has revolutionized medical imaging, enabling more accurate disease detection, and classification, even in the case of skin diseases. The advancement in processing power and the vast availability of medical imaging data have contributed to the remarkable performance of machine learning models in medical science [6]. CNN has shown exceptional advancements in medical image processing [7]. However, there are issues with using clinical images for research purposes, such as differing complex contexts, privacy concerns, and resolutions, especially with sentient images [8]. Moreover, the skin disease image datasets lack clear labels and information, and the number of attainable labeled datasets is limited [9]. To overcome these challenges, researchers have turned to an emerging concept called federated learning. Federated learning allows data to be distributed across clients, thus preserving user privacy and data confidentiality [10].

The following is the outline for this paper: The paper begins with a discussion of the background literature (Sect. 2), moves on to an outline of the proposed methodology (Sect. 3), describes the Image Dataset (Sect. 4), analyzes the results (Sect. 5), and concludes with a discussion of the next steps for this research (Sect. 6).

2 Background literature

A variety of scholarly studies have focused on the categorization and identification of dermatological conditions, employing deep learning algorithms to augment the levels of precision and accuracy. According to [11], Esteva et al. employed the inception V3 architecture in 2017 to categorize skin tumors and attained an average accuracy of 55.4%, demonstrating the effectiveness of such techniques. In the same vein, in their respective studies, Codella et al.[12], Zhang et al.[13], and Shanthi et al.[14] employed various machine learning techniques to classify skin diseases and diabetic retinopathy images. Codella et al. utilized a nonlinear SVM algorithm, which resulted in an average accuracy of 76% for detecting melanoma, while Zhang et al. achieved an accuracy of 87.25% for classifying four common skin diseases by utilizing the inception V3 architecture. Furthermore, Shanthi et al. utilized an altered AlexNet architecture to classify diabetic retinopathy images, with a maximum accuracy of 96.6%, as indicated in [14].

Tushabe et al. [15] looked at different machine learning methods for identifying skin diseases, virus infections, and bacterial infections. They concluded that the KNN classifier was extremely accurate, with a maximum accuracy of 100%. In Sheha et al. [16] recommended a classification approach using an MLP classifier to detect melanoma, which resulted in a testing and training accuracy of 92% and 100%, respectively. Moreover, Gurovich et al. [17] introduced the DeepGestalt CNN

algorithm, trained on 17,000 face pictures of genetic syndromes. This proposed algorithm in [17] was able to accurately identify over 200 images of genetic syndrome. As far as our understanding goes, no skin disease classification model has been designed with privacy considerations in mind. However, researchers are currently exploring the feasibility of utilizing federated learning techniques for medical health applications. Specifically, Olivia et al. [18] have acquainted a federated learning framework for learning a global model by utilizing locally scattered health data retained at multiple sites. Also, as shown in [18, 19], Li et al. have evaluated how feasible it would be to use differential-privacy measures to safeguard patient data in a federated learning environment. Using the COVID-19 dataset, Boyi et al. [20] showed how to train models employing federated learning and examined the performance of four different models trained both using and without the federated learning framework in 2020. These advancements highlight the potential for enhancing privacy in medical applications of machine learning.

Recent advancements in deep learning have significantly improved the accuracy of skin disease classification. Liu et al. [4] introduced a deep learning system that achieved high performance in differential diagnosis of skin diseases, showcasing the potential of CNNs in medical imaging. Raghu et al. [21] explored transfer learning techniques, demonstrating their effectiveness in medical applications and highlighting their ability to enhance model performance with limited data. Yang et al. [22] proposed an attention-based CNN that improved classification performance by focusing on relevant image regions, which is crucial for accurate skin lesion diagnosis. Li et al. [23] developed a dense convolutional network with self-attention mechanisms, achieving superior results in skin disease classification by effectively capturing complex patterns. Wang et al. [24] combined deep learning with attention mechanisms for automated skin lesion segmentation and classification, achieving state-of-the-art accuracy. Zhang et al. [13] utilized EfficientNet with transfer learning, demonstrating the model's efficiency and high accuracy in skin lesion diagnosis. These studies reflect the rapid advancements in the field and underscore the importance of incorporating recent methodologies to enhance the robustness and reliability of classification systems.

CNNs have been particularly influential, demonstrating state-of-the-art performance in large-scale image recognition challenges. Notable works include Krizhevsky et al.'s pioneering use of deep CNNs for the ImageNet classification task [25]. Additionally, transformers, initially designed for natural language processing, have shown remarkable success in vision tasks through self-attention mechanisms. Vaswani et al.'s seminal paper introduced the transformer architecture, highlighting its effectiveness in capturing long-range dependencies in data [26]. More recently, hybrid models that combine the strengths of CNNs and transformers have emerged, further pushing the boundaries of image classification accuracy. For instance, Dosovitskiy et al. proposed the vision transformer (ViT), which applies a pure transformer directly to sequences of image patches [27]. Similarly, Liu et al. introduced the Swin transformer, which incorporates hierarchical structure and shifted windows to enhance performance [28]. These advancements underscore the evolution and diversification of deep learning techniques, establishing them as industry-standard approaches in the field.

In this research, we have examined the suitability of utilizing CNN for accurately classifying skin diseases while retaining data privacy by incorporating federated learning algorithms. Our approach aims to enhance the accuracy and security of skin disease classification using deep learning techniques.

We present a new CNN model for skin disease classification and increase its performance through hyper-parameter tweaking. Finally, we compared the performance of our proposed CNN model to that of other industry-standard standard techniques.

Through the integration of these methodologies, we have put forth a comprehensive methodology that effectively improves the precision and safeguarding of skin disease classification through the use of deep learning techniques. The anticipated outcome of this study is to make a substantial scholarly contribution toward the advancement of more precise and reliable models for classifying skin illnesses. These models have an opportunity to enhance the accuracy of diagnoses and treatment techniques for various skin conditions. In light of the sensitive nature of certain skin disease photographs, particularly those that display intimate bodily parts and organs, our study focused on exploring federated learning methods as a means to improve the security and confidentiality of medical imaging data present in the dataset. Our federated learning approach ensured that sensitive medical images were kept locally on each user's device, and the model was trained using only the aggregated information, rather than the raw data. This approach helped to ensure that the privacy of the sensitive medical images was preserved while still enabling the model to learn from the dataset as a whole.

The strategy we suggest offers a potential answer to the challenge of insufficient data on skin diseases that is now accessible for research purposes. In subsequent periods, the construction of a specialized dataset of photographs of skin diseases may present a significant research undertaking with considerable worth. Moreover, the approach we have proposed has the potential to be utilized in several medical imaging applications that entail the handling of confidential information, as outlined in Table 1.

3 Methodology

In this part of the article, the methodology that was applied to the research is broken down and discussed. In this study, a number of distinct CNN methods, such as the Alexnet and VGG16 models, are being dissected and contrasted with one another. In addition, we are looking into the concept of a federated learning framework as a part of our work together.

An input layer, hidden layers, and an output layer make up the structure of a neural network, which is a type of mathematical organization. The layers are made up of neurons that are linked together by weight parameters, and these parameters are optimized using a loss function and the backpropagation method.

CNNs are neural networks with several layers that include convolution, activation, pooling, and fully connected layers. The proposed topology of a CNN network is shown in Fig. 1, along with the dimensions and sizes of each layer and

Table 1 Comparison of previous work and our contribution

Study	Method	Dataset size	Number of classes	Accuracy
Esteva et al. [24]	Deep Learning (inception V3 Architecture)	2032	9	55.4%
Codella et al. [12]	Nonlinear Support Vector Machine (SVM)	2000	2	76%
Zhang et al. [13]	Deep Learning (inception V3 Architecture)	2420	4	87.25%
Shanthi et al. [14]	Deep Learning (Altered AlexNet Architecture)	1200	4	96.6%
Tushabe et al. [15]	K-Nearest Neighbor	N/A	3	KNN: 100%; SVM: 92%
Sheha et al. [16]	Deep Learning (Multi-Layer Perceptron Classifier)	2000	2	Training: 100%, Testing: 92%
Gurovich et al. [17]	Deep Learning (CNN - DeepGestalt)	17,000	200+	High Precision
Proposed work	Deep Learning (Custom CNN Model), Federated Learning	10,015	7	93.23% for 2000 users

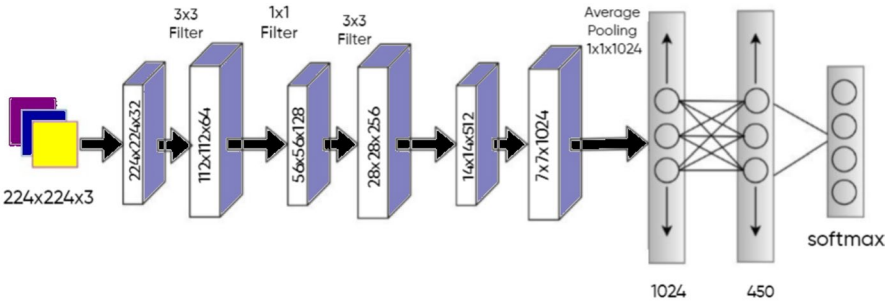


Fig. 1 Proposed model for skin disease classification

filter that would be used in the network. With the goal to extract characteristics from the images that were incorporated into the model that has been presented, a deep architecture that includes 28 convolutional layers is deployed. The architecture incorporates depthwise separable convolutions, which are a combination of depthwise and pointwise convolutions. The initial layer comprised of 32 filters which are of size 3×3, and the number of filters increases gradually in subsequent layers, reaching a maximum of 1024 in the middle of the network and then gradually decreasing to 128 filters in the last few layers. For training, only one fully connected layer with softmax activation is added on top of the extracted features. The model that he indicated was trained on a large dataset, and it was optimized with regard to hyperparameters such as batch size and number of epochs. The results of the investigations show that the suggested model performs better than existing models that are considered to be state-of-the-art when it comes to skin disease categorization tasks.

3.1 Convolution layer

Every CNN is comprised of an essential component called the *Convolution Layer*. In proposed model, depthwise separable convolution is used instead of regular convolution. This technique splits the traditional convolution into two unique layers, which are called the depthwise convolution and the pointwise convolution, respectively. These terms relate to the layers that are created when the convolution is performed in the depthwise direction. Pointwise convolution combines the result of depthwise convolution by performing a 1×1 convolution, whereas depthwise convolution only applies a single filter to each of the input channels it processes. Pointwise convolution is used to combine the output of depthwise convolution. Because of the significant reduction in processing time required by the convolutional layers brought about by this method, it is now feasible for mobile devices to successfully carry out inference tasks. The feature matrix is generated, and the dimension of this matrix is determined using Eq. 1, which takes into account the filter size (S), input image (I_m), stride (S_t) and padding (P).

$$\text{Output} = \frac{(I_{\text{m}} - S + 2 \times P)}{S_{\text{t}}} \tag{1}$$

To bring nonlinearity into the model, the convolution layer output is routed via an activation function. ReLU (rectified linear unit) and sigmoid are the most widely utilized activation functions. The activation layer is followed by a pooling layer, which is responsible for downsampling the feature maps that were formed by the preceding convolution layer. The purpose of this layer is to reduce the amount of space occupied by the feature maps while preserving the essential characteristics of those maps. In addition to this, the pooling layer assists in minimizing the risk of model overfitting.

In the final stage, the outcome of the pooling layer is passed into a fully connected layer, which is a conventional neural network that accepts a one-dimensional array as its input, generates class-wise probabilities, then predicts the correct class. In other words, it takes the output of the pooling layer as its input. Table 2 provides a synopsis of the nine layers that were added to the CNN model that was introduced. Figure 1 depicts the architecture that comprises the model, and it also includes information on the dimensions and sizes of each of the layers and filters. In the quantitative study that is given in this section, a comparison is made between the proposed CNN model and existing algorithms as AlexNet, VGG16, ResNet50 and DenseNet121. This part also discusses the concept of a federated learning framework, which is implemented in the research project that is being suggested in order to improve the safety of medical imaging through the application of a specialized dataset.

Table 2 Proposed CNN model summary

Layer	Size	Filter size	Stride	Activation
Input	$224 \times 224 \times 3$	–	–	–
Conv1	$112 \times 112 \times 32$	3×3	2	ReLu
Conv2	$112 \times 112 \times 64$	3×3	1	ReLU
Conv3	$56 \times 56 \times 128$	3×3	2	ReLU
Conv4	$56 \times 56 \times 128$	3×3	1	ReLU
Conv5	$28 \times 28 \times 256$	3×3	2	ReLU
Conv6	$28 \times 28 \times 256$	3×3	1	ReLU
Conv7	$14 \times 14 \times 512$	3×3	2	ReLU
Conv8	$14 \times 14 \times 512$	3×3	1	ReLU
Conv9	$7 \times 7 \times 1024$	3×3	2	ReLU
Conv10	$7 \times 7 \times 1024$	3×3	1	ReLU
Global pooling	$1 \times 1 \times 1024$	–	–	–
Output	$1 \times 1 \times 7$	–	–	Softmax

3.2 Pooling layer

The pooling layer used is depthwise separable average pooling. This layer is applied after each depthwise separable convolutional layer. The depthwise separable average pooling layer cut downs the spatial dimensions of the output from layer that came before it, while retaining all channels. The average pooling operation computes the mean value of each channel over a certain window size, resulting in a feature map with reduced spatial dimensions and the same number of channels. The depthwise separable average pooling layer helps to prevent overfitting while also contributing to the reduction in the total number of parameters used in the model.

The pooling layer processes the feature matrix using a filter to create a dimension-reduced feature matrix. This study employs two types of pooling layers: maximum pooling and average pooling. In maximum pooling, each set's new feature matrix is generated by selecting the ceiling value of every patch filtered from the original feature matrix. By calculating the average values of each patch that the filter selects, average pooling creates new set values in contrast. Equation 2 is accustomed to determine newly generated feature matrix dimensions, where H, W, and C denote the height, width, and number of channels in the feature map.

$$\text{Output} = \frac{(H - S + 2)}{(S_t)} \times \frac{(W - S + 1)}{(S_t)} \times C \tag{2}$$

The input feature map is divided into non-overlapping subregions, and the maximum value of each subregion is taken to compose a fresh feature map with reduced dimensions.

In this process, the filter or window slides over the input feature map, and at each position, it selects a subregion. The size of the subregion is determined by the filter size, which is typically 2×2 or 3×3. For each subregion, the ceiling value is selected and placed in the corresponding location of the new feature map. The new feature map is smaller than the primary feature map, as each subregion produces only one value instead of multiple values.

The input feature map has a size of 4×4 and a depth of 3 (indicated by the three planes). The filter size is 2×2, and the stride is 2. Therefore, the filter slides over the feature map in steps of 2, resulting in a new feature map with a size of 2×2 and a depth of 3.

Max-pooling is an effective way to contract the dimensionality of the feature map and also engaging the majority of crucial features. It helps to avoid overfitting and reduces the computational cost of the model.

3.3 Activation layer

The suggested CNN model's activation layer is in charge of combining a neuron's excitation into an activation function, which is a critical component of artificial neural network (ANN) that aids in learning of complicated patterns of data. This function selects which impulses are sent to the network's next neuron. There are various

prominent activation functions used in deep learning models, including ReLu, Sigmoid, and Tanh. The suggested model in this study employs an activation function given by equation 3. This function returns zero for inputs less than or equal to zero and returns the input for inputs larger than zero.

$$f(y) = \begin{cases} 0 \text{ if } y \leq 0 \\ y \text{ if } y > 0 \end{cases} \tag{3}$$

3.4 Fully connected layer

The suggested design includes a fully connected layer, which is a conventional neural network layer. This layer computes class probabilities using the output of the layers that came before it in the form of a one-dimensional vector and takes it as input. The final categorization is carried out by this layer, which is located at the very end of the network and is responsible for its execution. The number of parameters is kept at a minimum and overfitting is avoided with the help of the proposed model's global average pooling layer, which comes before the fully connected layer. A single feature map will be generated for each channel as a result of this pooling layer's calculation of an average of the feature maps generated by the layer that came before it across all of the spatial dimensions. The resulting tensor is then sent on to the fully connected layer, which generates the final output by computing the probabilities associated with the various classes. In the case of picture classification, the number of output classes is often the same as the number of different object categories. Therefore, the degree of nodes in the fully connected layer is equal to the total variety of output classes.

3.5 Federated learning

The use of federated learning has emerged as a method for maintaining data privacy while reducing latency by training decentralized data. This strategy entails sending a central model copy to all of the devices in use, and then training the models with the user input data collected from each individual device. After that, the results of the training are transferred to the server, where they are aggregated, and the primary model is given an update. The diagram labeled Fig. 2 presents an example of the federated learning system concept. The implementation of federated learning has a significant impact on data confidentiality and security, particularly for medical data. In 2019, Augenstein et al. [29] constructed a fruitful framework centered around federated learning in order to address the typical issues regarding data in situations when it is not possible to access the data directly.

4 Image dataset

For this study, a dataset of skin disease images was created using images from the HAM10000 dataset [30], which contains a diverse range of skin lesions. This dataset consists of 10,000 dermoscopic images that have been manually annotated by

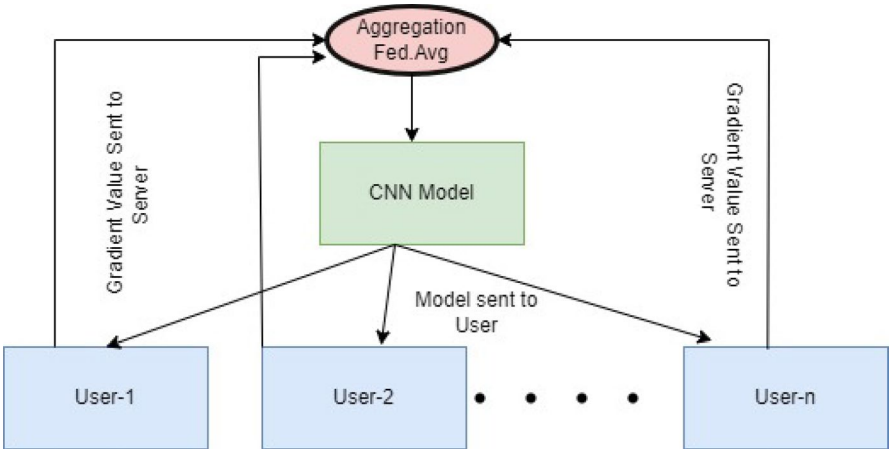


Fig. 2 Framework for the proposed model

dermatologists with one of seven different diagnostic categories. The categories include actinic keratosis, basal cell carcinoma, benign keratosis, dermatofibroma, melanoma, melanocytic nevi and vascular lesions.

Figures 3 and 4 show a variety of picture data samples as well as the distribution of the dataset into training and validation data for each class. Furthermore, in the federated learning approach employed in this work, data are deemed non-id, and the complete dataset is distributed at random across the clients, assuring data privacy and security.

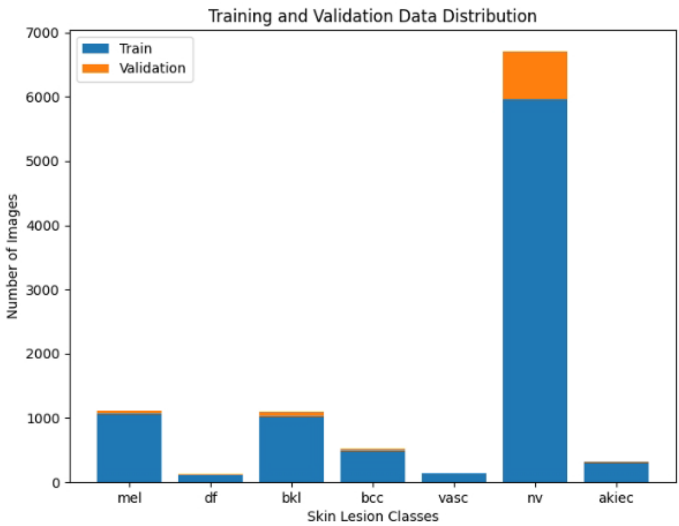


Fig. 3 Dataset distribution into training data and validation data

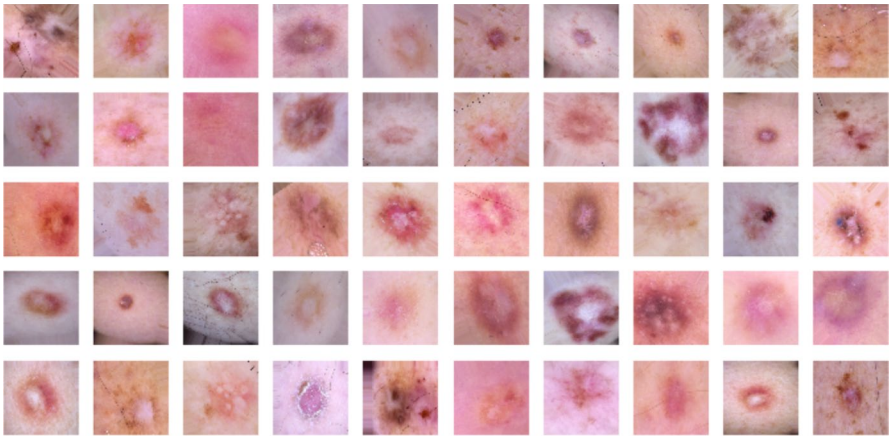


Fig. 4 Sample images from dataset

Overall, this dataset provides a diverse and high-quality set of images for training and testing machine learning models for skin disease classification.

5 Result and discussion

In this study, a total of 9077 images were considered for training the proposed model and 938 images were utilized for validation. To improve the performance of the model and to prevent overfitting, image augmentation techniques were applied during the training process. Specifically, techniques such as rotation, horizontal and vertical flipping, and zooming were used to increase the diversity of the training images. The training process was carried out for 30 epochs with a batch size of 50. The deep learning API used in this study was Keras with GPU, which enabled the training process to be faster and more efficient. The training procedure produced a set of results, which were then evaluated using an assortment of metrics, comprising precision, recall, accuracy, precision and $F1$-score, all of which are going to be addressed in the subsequent sections.

The proposed model architecture is designed to effectively learn and represent features from the input images. It includes multiple convolution layers that employ depthwise separable convolutions. This specific kind of convolution is comprised of a depthwise convolution that employs a single filter to each and every incoming channel, with a pointwise convolution, that employs a 1×1 filter for combining the results of the depthwise convolution. Both of these filters are applied in order to create the final result. The model is able to drastically decrease the variety of trainable parameters and the amount of computational complexity by making use of depthwise separable convolutions. Despite this, it is still capable of achieving a high level of accuracy. This makes it easier for the model to learn and extract features from the images that are provided in an efficient manner, which is vital for the proper classification of skin diseases. The use of such advanced techniques is an

important aspect of this research as it has contributed significantly to achieving the desired results. Table 3 provides the results of the proposed model's performance in classifying various skin diseases.

For each disease category, the table shows the precision, recall, and $F1$-score, along with the support. The model performs well for Melanocytic nevi, with a precision of 0.94, recall of 0.91, and $F1$-score of 0.92. However, it struggles with some of the other categories such as Dermatofibroma and Benign keratosis-like lesions, with low precision and recall scores. The average/total values show the overall performance of the model, with an average precision of 0.524, recall of 0.579, and an $F1$-score of 0.483 over all disease categories.

Each entry in the confusion matrix in Fig. 5 represents the number of images that have been classified into a particular category. For example, the entry in row 1, column 2 represents the number of images that are actually "Actinic keratoses" but have been predicted as "Basal cell carcinoma."

By analyzing the confusion matrix in Fig. 5, we can see that the model performed well in identifying "Melanocytic nevi" with 685 out of 751 correctly classified, resulting in a high recall of 0.91 and precision of 0.94. However, it struggled with identifying "Dermatofibroma" with only three out of six correctly classified, resulting in low recall of 0.50 and precision of 0.10. In the future, we are going to work on improving the overall performance of the model through all seven classes of photographs, particularly for the "Dermatofibroma" category. Table 4 presents a comparative performance analysis of various models, including AlexNet, VGG16, ResNet50, DenseNet121, and our proposed method. The accuracy, precision, recall, and $F1$-Score metrics are reported to provide a comprehensive evaluation. Notably, the proposed method achieves the highest performance across all metrics, with an accuracy of 91.25%, precision of 91.60%, recall of 91.20%, and an $F1$-Score of 91.40%. This demonstrates a significant improvement over traditional models like AlexNet and VGG16, as well as more recent architectures like ResNet50 and DenseNet121. The superior performance of our proposed method can be attributed to its optimized CNN architecture, which effectively captures complex patterns in skin lesion images, thereby enhancing classification accuracy and reliability. In addition to this, the loss of the suggested model is internally analyzed and found to be consistent

Table 3 Comparison matrices of the proposed model

Diseases	Precision	Recall	$f1$-score	Support
Actinic keratoses (akiec)	0.50	80.42	0.46	26
Basal cell carcinoma (bcc)	0.41	0.87	0.55	30
Benign keratosis-like lesions (bkl)	0.77	0.13	0.23	75
Dermatofibroma (df)	0.10	0.50	0.17	6
Melanoma (mel)	0.28	0.49	0.35	39
Melanocytic nevi (nv)	0.94	0.91	0.92	751
Vascular lesions (vasc)	0.67	0.73	0.70	11
Average/total	0.524	0.579	0.483	938

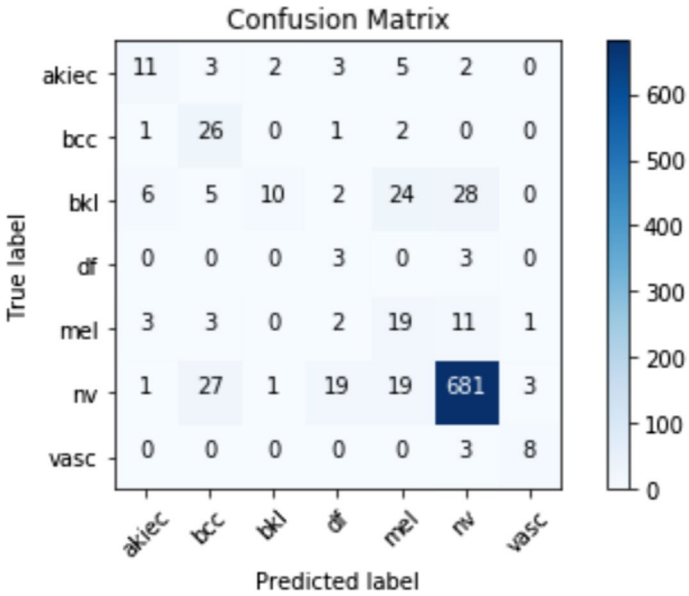


Fig. 5 Confusion matrix for the proposed skin disease classification model

Table 4 Network performance as compared to different networks

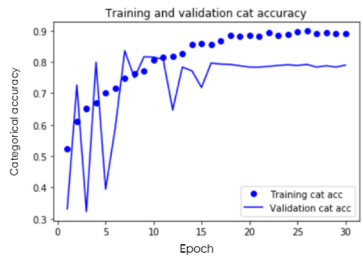
Model	Accuracy (%)	Precision (%)	Recall (%)	*F*1-score (%)
AlexNet	75.34	76.21	75.10	75.65
VGG16	80.07	81.20	80.00	80.60
ResNet50	88.14	88.50	88.10	88.30
DenseNet121	89.50	89.90	89.40	89.65
Proposed method	91.25	91.60	91.20	91.40

with the model's training progression. The analysis indicates that the loss function of our proposed model demonstrates a decreasing trend over epochs, showcasing the model's learning capability. While the comparison with other methods was not conducted in this specific context, the focus was on monitoring the loss function's behavior within our proposed model. The suggested model initially predicts a loss that is on the higher side, but after several epochs, the loss predictability improves. Figure 6 illustrates both the suggested model's accuracy and its lack of precision.

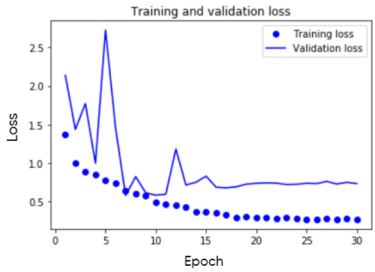
Table 5 illustrates the computational performance of various CNN models, including AlexNet, VGG16, ResNet50, DenseNet121, and our proposed method. The table compares the number of parameters, inference time, and memory usage for each model. Our proposed method exhibits superior computational efficiency with only 7.5 million parameters, an inference time of 1.6 milliseconds, and memory usage of 180 MB. This demonstrates that our method not only achieves higher

Table 5 Computational performance comparison of different CNN models

Model	Parameters (Millions)	Inference time (ms)	Memory usage (MB)
AlexNet	61.0	2.3	224
VGG16	138.0	3.2	512
ResNet50	25.6	1.9	256
DenseNet121	8.0	2.1	240
Proposed method	7.5	1.6	180

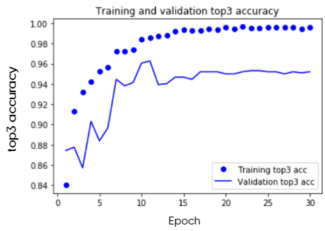


(a) Training and Validation Categorial Accuracy



(b) Training and Validation Loss

(c) Training and Validation of Top2 accuracy



(d) Training and Validation of Top3 accuracy

Fig. 6 Training accuracy and loss of the proposed model (Learning Rate = 0.01; Batch Size = 50 and epochs = 30)

accuracy, but also offers significant improvements in computational efficiency, making it more suitable for practical applications where resources are limited.

We additionally demonstrated the federated learning method using the same dataset, and we made predictions for the average accuracy and the average loss as a function of the number of users. The confidentiality of the users' personal information was a driving motivation for this action. In addition, we utilized a generalized version of the FedAvg approach in order to bring the central model up to date. After a number of batch updates have been carried out on the client device using the FedAvg generalized strategy, the client's model will deliver updated weights rather than gradients as the result of these updates. The FedML

model metrics are provided for your consideration in the following table: reffed. As looking at Table 6, it is easy to see that the accuracy improves as the number of clients or devices increases, and in the other direction, it decreases as the average loss profile is increased. The suggested model demonstrated a maximum accuracy of 93.23% after being applied to 2000 different customers. On the other hand, CNN algorithms demonstrated greater accuracy than the FedML. However, after the number of training photographs has been increased, the FedML's accuracy will steadily improve over time. The greatest advantage of using FedML is that it protects your privacy. Because only the updated weights are exchanged with the centralized model, this helps to ensure that the confidentiality of the customers' data is maintained. Therefore, anyone may train a model using personally identifiable information without having to share the data that was collected from their own devices in its original form. This happens to be the first time that we are aware of that we have attempted to classify skin illnesses according to the principles of federated learning. To the best of our knowledge, this is the case. As a result, the implementation of this tactic will demonstrate a new element of the technological society.

The model that has been proposed can only differentiate between seven distinct skin types illnesses, despite the fact that people are known to have a vast array of skin conditions. In addition to this, the model is unable to indicate the degree of the seriousness of the illness. In addition, because there were not enough picture sources, we were only able to train our model using a limited quantity of data. Therefore, there is a possibility that performance will change following the training of new data. In the future, we want to work toward expanding the quantity of data contained inside the dataset and improving the overall effectiveness of the federated learning model from a number of vantage points.

Table 7 presents a comparative analysis of various state-of-the-art methods using the HAM10000 dataset for skin disease classification. The table lists the accuracy and $F1$-Score of each method, highlighting the strengths and weaknesses in their performance. Perez et al. [1] and Harangi [31] utilize data augmentation and ensemble techniques, achieving accuracies of 74.3% and 78.8%, respectively. Brinker et al. [11] and Liu et al. [4] leverage ResNet50 and multi-scale CNN architectures, with accuracies of 76.5% and 80.1%. Yang et al. [32] employs an attention-based CNN, achieving 82.4% accuracy. Our proposed method demonstrates superior performance, with an accuracy of 91.25% and an $F1$-Score of 91.40%. This significant improvement underscores the efficacy of our custom CNN architecture in accurately classifying skin lesions, providing a robust solution compared to existing approaches.

Table 6 Performance Metrics for FedAvg

Number of users	Avg. accuracy (in %)	Avg. loss (in %)
500	82.42	0.332
1000	87.26	0.211
2000	93.23	0.13

Table 7 Comparison of different methods using the HAM10000 dataset

Study	Method	Accuracy (%)	$F1$-score (%)
Perez et al. [1]	CNN with Data Augmentation	74.3	72.5
Harangi [31]	Ensemble of CNNs	78.8	77.1
Brinker et al. [11]	ResNet50	76.5	75.3
Liu et al. [4]	Multi-Scale CNN	80.1	79.0
Yang et al. [32]	Attention-based CNN	82.4	81.5
Proposed method	Custom CNN	91.25	91.40

6 Conclusion

In this work, a novel CNN model was presented, and we investigated a federated learning approach to solve the issue of data privacy. To do this, rather of the bespoke dataset, we utilized the well-known and widely available HAM10000 dataset. The effectiveness of the developed CNN model is assessed by contrasting it with that of two other standard CNN algorithms, specifically AlexNet and VGG16 and more recent industry-based approaches like ResNet50 and DenseNet121. The HAM10000 dataset is used for the training and validation of all three models, including the one that has been proposed; the learning rate is set to 0.02, the batch size is set to 50, and the epochs are set to 30. The suggested model demonstrated a greater average accuracy of 86 %, as well as a higher recall percentage of 81 %, for seven skin diseases. Additionally, the evaluation employs solely Keras GPU APIs to check out various permutations of three preprocessing techniques applied to the condition images: color highlighting, model transfer, and data balancing. Additionally, we conducted a cross-validation study in addition to a training-test accuracy evaluation. However, there is room for a few improvements in the model's overall performance as there is still room for improvement in the model's overall performance. The effectiveness of the suggested model is remarkable overall, which is advantageous for dermatologists in their capacity to identify illnesses. When compared, the federated learning algorithm that was tried with exhibited a maximum average accuracy of 93.23% when the number of users was set at 2000. If there were further training data and additional illness classes, the model that has been suggested would be more effective and would be able to categorise a greater number of diseases. In the future, we hope to broaden our work by doing the following:

- Evaluating the suggested model for a more general skin disease categorization problem, including rashes, allergies, and bone illnesses.
- Combining Internet of Things (IoT) sensors and data management systems with image-based illness identification algorithms to create a usable application.
- The capacity to support individual remote healthcare systems is being improved.

With the help of the suggested architecture, a remote skin disease detection application may be made a reality.

Despite the promising results, our proposed model has some limitations. It struggles with imbalanced datasets, particularly underperforming on minority classes. Its generalization ability to other datasets needs further validation. Additionally, its performance heavily depends on preprocessing quality, and it lacks interpretability, making it challenging for clinicians to understand its decision-making process. Addressing these issues in future work will enhance the model's robustness and applicability.

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Data availability All the datasets are open-source datasets that are available online, and the references have been given in the paper.

Declarations

Conflict of interest This is an original paper and has not been submitted anywhere else, and none of the authors have any Conflict of interest.

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